Inference for Identified Sets via Classification-Based Metropolis-Hastings

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Insper

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1. Introduction

2. Setup and Main Idea

3. Theoretical Framework
   Some results

4. Simulation

5. Conclusion

Inference under partial identification in models with intractable likelihood

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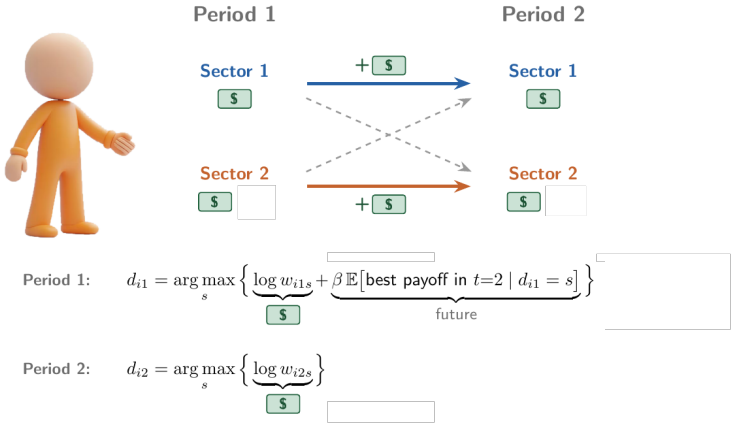
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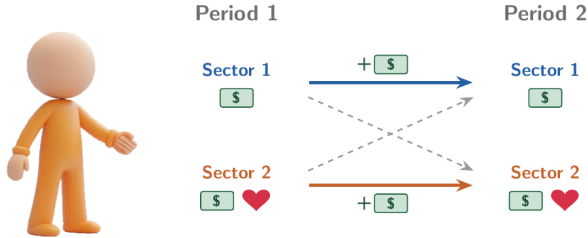
- Point identified structural models are usually estimated by maximum likelihood (MLE). In many of them the **likelihood cannot be evaluated** (no closed form, and no reliable numerical approximation).
- Also, it is often difficult to verify whether parameters in complicated nonlinear structural models are **point identified**.
- Each difficulty has its own literature. Usually, they do not overlap.
- **This paper:** confidence sets (CS) for the identified set in models that can be simulated but whose likelihood cannot be evaluated (without falling back on restricted features of the data distribution).

Example: a dynamic Roy model



- **Point identification** requires that workers choose purely on money.

Example: a generalized dynamic Roy model



- A worker who switches loses welfare because **experience** does not transfer. She can also lose in the case that her **unobserved ability** does not transfer.

Policy question: Should policy fund retraining or subsidize the move? (Dix-Carneiro, 2014)

Example: a generalized dynamic Roy model

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Help

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- **The likelihood is not evaluable:** when the unobservables states are serially correlated the likelihood is intractable ($\rho_t \neq 0$) (KMP, 2023).
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  - Evaluating the likelihood means integrating over the **entire history** of unobserved states.
- **But it is simulable:** draw shocks Z_j, from a known distribution P_Z. For any θ, T_θ solves the worker's two-period problem and returns

$$X_{j,\theta} = T_\theta(Z_j) = \left(d_{j1},\, \log w_{j1d_{j1}},\, d_{j2},\, \log w_{j2d_{j2}}\right)$$

(KMP, 2023).

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Example: a generalized dynamic Roy model

- **The model is partially identified:** the cross sector correlation (comparative advantage) is not point identified (Vijverberg, 1993).
  - If $V = 0$ we can use **switchers** to point identify ρ_{12}.
- **The likelihood is not evaluable:** when the unobservables states are serially correlated the likelihood is intractable ($\rho_t \neq 0$) (KMP, 2023).
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The question is: *how can we build a CS for the identified set here?*

Other applied works

- **Labor** - Taber and Vjelin (2020); Dix-Carneiro (2014)
- **Education** - Mourifié, Henry and Méango (2020); Todd and Wopin (2006)
- **IO** - Chen, Christensen and Tamer (2018)

Methodological Literature Review

- Estimation and inference for intractable likelihood models: Kaji, Manresa & Pouliot (KMP, 2023); Kaji & Ročková (KR, 2023).
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- Bayesian CS with valid frequentist coverage: Chen, Christensen & Tamer (CCT, 2018).
  - *Contribution:* extend to intractable likelihoods, without falling back on moments or **restricted features** of the data distribution.

Methodological Literature Review

- Intractable likelihood models, but simulable: Kaji, Manresa & Pouliot (KMP, 2023); Kaji & Ročková (KR, 2023).
  - *Contribution:* extend to partially identified models.

- Bayesian Partial Identification with frequentist coverage: Chen, Christensen & Tamer (CCT, 2018).
  - *Contribution:* extend to intractable likelihoods, without falling back on moments or restricted features of the data distribution.

This paper's contribution: asymptotically valid frequentist CS for the identified set, in models that can be simulated but whose likelihood is intractable.

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Setup

- $\mathcal{P} = \{p_\theta : \theta \in \Theta\}$ (not evaluable, but simulable), $\Theta \subseteq \mathbb{R}^k$;
- Correctly specified ($p_0 \in \mathcal{P}$), with $X_1, \ldots, X_n \overset{iid}{\sim} P_0$.

Main idea

1 - *I start building on the CCT (2018) framework:*

- The identified set is where the **expected log-likelihood** $\big(L(\theta) = E[\log(p_\theta(X))]\big)$ is maximised. Under correct specification the maximum is known: $\max_\Theta L = 0$. So the identified set is the zero level set

$$\Theta_I \;=\; \{\theta : -D_{\mathrm{KL}}(p_0 \| p_\theta) = 0\} \;=\; \{\theta : p_\theta = p_0 \text{ a.e.}\},$$

  the observational-equivalence class of p_0 (Cover & Thomas, 2006).

- Sample analogue: $L_n(\theta) = \frac{1}{n}\sum_i \log p_\theta(X_i)$.

- The CCT's quasi-likelihood ratio statistic and the posterior

$$Q_n(\theta) := 2n\big[L_n(\theta^*) - L_n(\theta)\big], \qquad \Pi_n(d\theta) \;\propto\; e^{nL_n(\theta)}\,\Pi(d\theta),$$

  where Π is a prior on Θ and $\theta^* \in \Theta$ is any approximate maximizer of L_n.

- CCT's CS for the Θ_I is $\widehat{\Theta}_\alpha = \big\{\theta \in \Theta : Q_n(\theta) \leq \xi^{\text{mc}}_{n,\alpha}\big\}$, where $\xi^{\text{mc}}_{n,\alpha}$ is the α-quantile of $\{Q_n(\theta^1), \ldots, Q_n(\theta^B)\}$ and $\theta^1, \ldots, \theta^B$ are draws from the posterior Π_n.

Question: How can we construct this CS if we cannot evaluate the likelihood?

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Question: How can we construct a CS if we cannot evaluate the likelihood?

2 - *KR (2023) has the solution:*

- Replace the infeasible log-likelihood $L_n(\theta)$ by the classification-based log-likelihood to get the **QLR**.
- Use the **classification-based MH algorithm** to sample from the quasi-posterior.

What need to be shown?

1. The CS built from the classifier-based likelihood retains asymptotically valid **frequentist** coverage for Θ_I.
   - **Need to be proved:** the classifier-based QLR has the same asymptotic behaviour as the infeasible QLR.
   - **What it takes:** the classifier estimator converges to the true, uniformly at a rate standard nonparametric estimators attain.

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2. KR's classification-based MH algorithm works in partially identified settings. (**For the future**)

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The likelihood is simulable but not evaluable

Assumption 1: [Intractable likelihood and simulable model] For each $\theta \in \Theta$, $p_\theta(x)$ is **intractable**. Hence $L_n(\theta)$ is unavailable.

But the model can be **simulated**. Draw shocks $Z_1, \ldots, Z_m$ from a known distribution P_Z and push them through the structural map:

$$X_{j\theta} \;=\; T_\theta(Z_j), \qquad j = 1, \ldots, m$$

(KMP, 2023).

The cross-entropy classifier

- The **likelihood ratio** can be recovered from the cross-entropy classifier solution (KR, 2023).

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- The population maximiser of the cross-entropy objective

$$M_\theta(D) \;=\; P_0\big[\log D\big] \;+\; P_\theta\big[\log(1-D)\big], \qquad \text{for any } D : \mathcal{X} \to [0,1]$$

is given by the **oracle classifier**

$$D_\theta(X) \;:=\; \frac{p_0(X)}{p_0(X)+p_\theta(X)} \quad\Longrightarrow\quad \frac{p_\theta}{p_0} = \frac{1-D_\theta}{D_\theta}, \qquad \text{(Goodfellow et al., 2014)} \;\; (1)$$

It is **estimated** by its empirical counterpart over a sieve $\mathcal{D}$, with the observed data as one class and the simulated data as the other (KR eq. 2.2):

$$\widehat{D}^{\theta}_{n,m} \;=\; \underset{D\in\mathcal{D}}{\arg\max}\; \frac{1}{n}\sum_{i=1}^{n} \log D(X_i) \;+\; \frac{1}{m}\sum_{j=1}^{m} \log\big(1-D(X_j^\theta)\big).$$

Likelihood estimation with a classifier

The *classifier-based log-likelihood*

$$\widetilde{L}_n(\theta) := L_n(\theta) + \frac{1}{n} u_\theta(X^{(n)}),$$

where $u_\theta(X^{(n)}) := \sum_{i=1}^{n} \left[\log \frac{1-\widehat{D}^{\theta}_{n,m}(X_i)}{1-D_\theta(X_i)} - \log \frac{\widehat{D}^{\theta}_{n,m}(X_i)}{D_\theta(X_i)} \right]$ (KR eq. 2.5).

KR's classifier likelihood is not orthogonal in D

At a **generic** classifier D, the population counterpart of KR's classifier-based log-likelihood is

$$\Psi(\theta, D) \;:=\; P_0\left[\log \frac{1-D}{D}\right]$$

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At a **generic** classifier D, the population counterpart of KR's classifier-based log-likelihood is

$$\Psi(\theta, D) \;:=\; P_0\left[\log \frac{1-D}{D}\right]$$

Proposition 1 [Non-orthogonality] For every $\theta \in \Theta$ and every bounded direction Δ,

$$\left.\frac{\partial}{\partial t}\right|_{t=0} \Psi(\theta, D_\theta + t\Delta) = -\,P_0\left[\frac{\Delta}{D_\theta(1-D_\theta)}\right],$$

The problem. We would need $\widehat{D}^{\theta}_{n,m}$ to converge to the oracle **faster than the parametric rate**. No estimator delivers this.

Likelihood estimation with a classifier

The orthogonalized classifier-based log-likelihood

$$\widetilde{L}_n(\theta) := L_n(\theta) + \frac{1}{n} u_\theta(X^{(n)}) + \left(\frac{1}{m} c_\theta(X_\theta^{(m)}) - 1 \right) \tag{2}$$

where $u_\theta(X^{(n)}) := \sum_{i=1}^{n} \left[\log \frac{1-\widehat{D}^\theta_{n,m}(X_i)}{1-D_\theta(X_i)} - \log \frac{\widehat{D}^\theta_{n,m}(X_i)}{D_\theta(X_i)} \right]$ and $c_\theta(X_\theta^{(m)}) := \sum_{j=1}^{m} \frac{\widehat{D}^\theta_{n,m}(X_j^\theta)}{1-\widehat{D}^\theta_{n,m}(X_j^\theta)}$

At the oracle, the population substitution preserves the target

Proposition 2 [Target preservation]

$$\widetilde{L}(\theta) \;=\; L(\theta) \;=\; -D_{\mathrm{KL}}\big(p_0 \,\|\, p_\theta\big) \qquad \text{for every } \theta \in \Theta,$$

hence $\arg\max_{\theta\in\Theta} \widetilde{L}(\theta) = \arg\max_{\theta\in\Theta} L(\theta) = \Theta_I$.

▸ Proof

The classifier-based QLR and quasi-posterior

Define the classifier-based quasi-likelihood ratio (QLR) and quasi-posterior

$$\widehat{Q}_n(\theta) := 2n\big[\widetilde{L}_n(\theta^*) - \widetilde{L}_n(\theta)\big], \qquad \widehat{\Pi}_n(d\theta) \;\propto\; e^{n\widetilde{L}_n(\theta)}\,\Pi(d\theta), \tag{3}$$

where Π is a prior on Θ and $\theta^* \in \Theta$ is any approximate maximizer of $\widetilde{L}_n$.

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where Π is a prior on Θ and $\theta^* \in \Theta$ is any approximate maximizer of $\widetilde{L}_n$.

Lemma 1 [Intractable terms disappear] For any $\theta, \theta' \in \Theta$,

$$n\widetilde{L}_n(\theta) - n\widetilde{L}_n(\theta') = \sum_{i=1}^{n}\left[\log\frac{1-\widehat{D}^{\theta}_{n,m}(X_i)}{\widehat{D}^{\theta}_{n,m}(X_i)} - \log\frac{1-\widehat{D}^{\theta'}_{n,m}(X_i)}{\widehat{D}^{\theta'}_{n,m}(X_i)}\right] + \frac{n}{m}\sum_{j=1}^{m}\left[\frac{\widehat{D}^{\theta}_{n,m}(X^{\theta}_j)}{1-\widehat{D}^{\theta}_{n,m}(X^{\theta}_j)} - \frac{\widehat{D}^{\theta'}_{n,m}(X^{\theta'}_j)}{1-\widehat{D}^{\theta'}_{n,m}(X^{\theta'}_j)}\right]$$

Important: $\widehat{Q}_n$ and the MH acceptance ratio is computable from $\{\widehat{D}^{\theta}_{n,m}\}_\theta$ alone (KR, 2023).

▸ Proof

CS for the Identified Set Θ_I

PROCEDURE (FOLLOWING CCT, PROCEDURE 1):

1. Draw $\{\theta^1, \ldots, \theta^B\}$ from the quasi-posterior $\widehat{\Pi}_n$ in (3).
2. Let $\widehat{\xi}^{\text{mc}}_{n,\alpha}$ be the α quantile of $\{\widehat{Q}_n(\theta^1), \ldots, \widehat{Q}_n(\theta^B)\}$.
3. The $100\alpha\%$ CS for Θ_I is

$$\widehat{\Theta}_\alpha = \left\{\theta \in \Theta : \widehat{Q}_n(\theta) \leq \widehat{\xi}^{\text{mc}}_{n,\alpha}\right\}. \tag{4}$$

Question: $\widehat{Q}_n$ is built from a classifier based likelihood. Does $\widehat{\Theta}_\alpha$ still have asymptotically valid frequentist coverage?

▸ Subvectors

A shrinking neighborhood for Θ_I

Definition 1: [Shrinking neighborhood for Θ_I] With $\varepsilon_n \to 0$ and $\varepsilon_n >> n^{-1/2}$ $\left(\text{e.g. } \varepsilon_n = n^{-1/2}(\log n)^{1/2}\right)$,

$$\Theta_{osn} \;:=\; \left\{\theta \in \Theta :\; h(p_\theta, p_0) < \varepsilon_n\right\}$$

Then $\Theta_I \subseteq \Theta_{osn}$ and $\Theta_{osn} \downarrow \Theta_I$.

Assumption 2: [Posterior contraction] Under regularity conditions (Ghosal, Ghosh, and van der Vaart (2000))

$$\widehat{\Pi}_n\left(\Theta_{\text{osn}}^c \mid X^{(n)}\right) = o_P(1),$$

where $\Theta_{\text{osn}}^c = \Theta \setminus \Theta_{osn}$.

Uniform flatness on Θ_{osn}

Theorem 1 *(proof in progress).* If $\delta_n = o\big(n^{-1/4-w}\big)$ for some $w > 0$ (e.g. $\delta_n = n^{-1/4}(\log n)^{-1/4}\big)$ and $m \gg n^{3/2}$, then

$$\sup_{\theta,\theta' \in \Theta_{osn}} \big|\big(u_\theta + c_\theta\big) - \big(u_{\theta'} + c_{\theta'}\big)\big| \;=\; o_P(1),$$

where $u_\theta + c_\theta = u_\theta(X^{(n)}) + c_\theta(X_\theta^{(m)}) := n\widetilde{L}_n(\theta) - nL_n(\theta)$.

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Implication: In the difference, the classifier-based likelihood reproduces the intractable one up to $o_P(1)$

$$n\big[\widetilde{L}_n(\theta) - \widetilde{L}_n(\theta')\big] = n\big[L_n(\theta) - L_n(\theta')\big] + o_P(1).$$

Main theorem

Theorem 2 [Valid frequentist coverage] Let Theorem 1 result hold. Then $\widehat{Q}_n(\theta) = Q_n(\theta) + o_P(1)$ *uniformly on* Θ_{osn}, and CCT's valid coverage theorem apply verbatim to $\widehat{Q}_n$, that is, $\liminf_{n\to\infty} \mathbb{P}\left(\Theta_I \subseteq \widehat{\Theta}_\alpha\right) \geq \alpha$.

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Simulation evidence: entry game example

- Each player simultaneously chooses whether to enter: $Y_j \in \{0, 1\}$.
- Payoffs depend on standalone profitability β_j, the competitive effect Δ_j, and the private shock ϵ_j.
- $\theta = \{\beta_1, \Delta_1, \beta_2, \Delta_2, \rho, s\}$

		Player 2	
		0	1
Player 1	0	$(0, 0)$	$(0, \beta_2 + \epsilon_2)$
	1	$(\beta_1 + \epsilon_1, 0)$	$(\beta_1 + \Delta_1 + \epsilon_1,\ \beta_2 + \Delta_2 + \epsilon_2)$

First entry in each cell is player 1's payoff; second is player 2's.

Algorithm: SMC (CCT, 2018) with Classification-based MH (KR, 2023)

For each MC repetition, draw $Z = (Z_1, \ldots, Z_m)$ once. Then at every θ:

- **Simulation** $\quad \widetilde{X}_j^\theta = T_\theta(Z_j), \quad j = 1, \ldots, m$
- **Classification problem** $\quad \widehat{D}^\theta \leftarrow$ real $X^{(n)}$ vs simulated X^θ
- **Evaluate the classifier likelihood** $\quad \widetilde{L}_n(\theta)$

Hand $\widetilde{L}_n$ to the sampler SMC + the classification based MH.

▸ Algorithm

Simulation evidence: implementation

Table: Entry game with classifier-based CS. $R = 500$, $n = 1{,}000$, $m = 5{,}000$, $B = 1000$, $\alpha = 0.95$, Projection method.

	Classifier-based (this paper) Avg. CS	CCT, Projection Avg. CS
Coverage for Θ_I	0.96	0.96
Δ_1	$[-1.52,\ 0.00]$	$[-1.54,\ 0.00]^{a}$
β_1	$[-0.13,\ 0.74]$	$[-0.14,\ 0.76]^{a}$
ρ	$[0,\ 1]$	$[0,\ 1]^{b}$
s	$[0,\ 1]$	$[0,\ 1]^{b}$
Δ_2, β_2	omitted; identical by symmetry	

(a) Identified sets for Δ_1 and β_1 are $[-1.42, 0.00]$ and $[0.05, 0.66]$.
(b) Identified set for ρ, s is $[0, 1]$

◂ QQ Plot

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Conclusion

- We were able to derive CS with valid frequentist coverage for the idenfied set even when the likelihood is intractable.

- Future Work:
  - Finish the theoretical results.
    - Drop the correct specification assumption.
  - Improve CS tightness with the Profile Criterion from CCT (2018).
  - Build the theory on the sampling layer.
    - How to better draw from the posterior (CCT, 2018; Kitagawa & Kuang, 2026).
  - Applied example.

Thank you.

The cross-sector correlation is not identified

- **Without taste, people move only for money.** Everything that drives the choice also shows up in a wage, so the data reveal how large the unobserved differences are. The movers' premium is then pure comparative advantage, and the correlation is pinned down.
- **With taste, a mover may be moving because she *likes* the sector**, not because she is good at it. Her premium looks the same either way. The data see one number where there are two stories: **ability that does not transfer**, or **taste that happens to line up with ability**.

Back

Why the likelihood is intractable

- **Serially correlated unobserved states.**
- The shocks $I_{it} = (\varepsilon_{it1}, \varepsilon_{it2}, V_{it})$ are persistent ($\rho_t \neq 0$) and unobserved by the econometrician.
- This breaks conditional independence, the assumption that keeps dynamic discrete choice likelihoods tractable.

Simulating the model: $X_{i,\theta} = T_\theta(Z_i)$

- **Draw shocks once:** $Z_i \sim U[0,1]^6$, one for each of $(\varepsilon_{i11}, \varepsilon_{i12}, V_{i1}, \varepsilon_{i21}, \varepsilon_{i22}, V_{i2})$.
- T_θ **turns shocks into data:**
  - Map Z_i into normal shocks with covariance $\Omega(\theta)$.
  - *Period 1:* the agent forecasts period 2 and picks the sector with the highest current wage + taste + present value of future welfare
  - *Period 2:* she picks the sector with the highest wage + taste.
- **Output:** $X_{i,\theta} = (\log w_{i1},\ d_{i1},\ \log w_{i2},\ d_{i2})$.

CS for Subvectors: Projection

Let $\mu : \Theta \to \mathbb{R}$ be the subvector of interest (e.g. $\mu(\theta) = \theta_k$, a single coordinate).

PROCEDURE (FOLLOWING CCT):

1. Reuse the draws $\{\theta^1, \ldots, \theta^B\}$ and the *same* critical value $\widehat{\xi}^{\text{mc}}_{n,\alpha}$ from the previous slide. No new quantile is computed.
2. Keep the draws inside the set: $\mathcal{B}_\alpha = \left\{b \leq B : \widehat{Q}_n(\theta^b) \leq \widehat{\xi}^{\text{mc}}_{n,\alpha}\right\}$.
3. The $100\alpha\%$ CS for $\mu(\Theta_I)$ is the *projection*

$$\widehat{M}_\alpha = \left[\min_{b \in \mathcal{B}_\alpha} \mu(\theta^b), \ \max_{b \in \mathcal{B}_\alpha} \mu(\theta^b) \right].$$

Coverage is inherited: $\Theta_I \subseteq \widehat{\Theta}_\alpha$ implies $\mu(\Theta_I) \subseteq \mu(\widehat{\Theta}_\alpha)$, so $\widehat{M}_\alpha$ covers $\mu(\Theta_I)$ with probability *at least* α.

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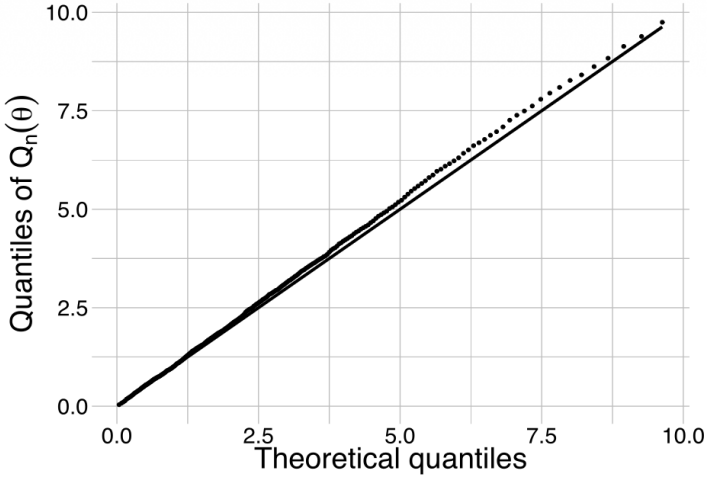


Figure: Reproduced from Fig. 2 (CCT, 2018)

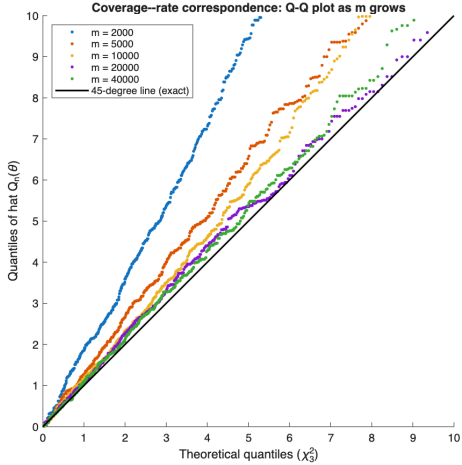


Figure: Q–Q plots of $\widehat{Q}_n(\theta)$ against χ^2_3 for $m \in \{2{,}000, 5{,}000, 10{,}000, 20{,}000, 40{,}000\}$.

Density constancy

Lemma 2 [Density constancy and near-constancy] Under correct specification, for every $\theta \in \Theta$, $D_{\mathrm{KL}}(p_0 \| p_\theta) \geq 0$, with equality iff $p_\theta = p_0$ (Thomas and Cover, 2006).

(i) Hence $p_\theta = p_0$ for every $\theta \in \Theta_I$, and in particular $p_\theta = p_{\theta'}$ for all $\theta, \theta' \in \Theta_I$.

(ii) Hence the oracle classifier is constant on the identified set:

$$D_\theta(X) \;=\; \frac{p_0(X)}{p_0(X) + p_\theta(X)} \;\equiv\; \tfrac{1}{2} \qquad \text{for every } \theta \in \Theta_I.$$

To be continued...

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Assumption 2: [Posterior contraction]

$$\widehat{\Pi}_n\left(\Theta_{\text{osn}}^c \mid X^{(n)}\right) = o_P(1),$$

where $\Theta_{\text{osn}}^c = \Theta \setminus \Theta_{osn}$.

Observation about assumption 2: regularity conditions still need to be proved (Theorem 5.1 of Ghosal, Ghosh, and van der Vaart (2000)).

... Continuation

Lemma 2 [Near-constancy of the oracle classifier] We know that $d\big(D_\theta, \frac{1}{2}\big) = h(p_\theta, p_0)$ for all $\theta \in \Theta$. By Definition 1, $h(p_\theta, p_0) < \varepsilon_n$ for every $\theta \in \Theta_{osn}$, hence

$$\sup_{\theta \in \Theta_{osn}} d\big(D_\theta, \tfrac{1}{2}\big) \;\leq\; \varepsilon_n.$$

▸ Proof

Observation: all the uniformity results are build on Θ_{osn} (following CCT, 2018)

The classifier converges uniformly over Θ_{osn}

Lemma 3: [Uniform classifier rate] Under a bracketing-entropy condition on the sieve,

$$\sup_{\theta \in \Theta_{osn}} d\big(\widehat{D}^{\theta}_{n,m}, D_{\theta}\big) \;=\; O_P(\delta_n)$$

Observation: Similar to KMP (2023) and KR (2023). The only change is that the entropy is computed for a *single aggregated class* $\bigcup_{\theta \in \Theta_{osn}} (\cdot)$

▸ Bracketing condition

Uniform bracketing entropy

Assumption 3: [Uniform bracketing entropy over Θ_{osn}]

With $\mathcal{F}_n := \bigcup_{\theta \in \Theta_{osn}} \left\{ \ell_D - \ell_{D_\theta} \ : \ D \in \mathcal{D}^{\theta}_{n,\delta_n} \right\}$, the rate δ_n is the largest solution of

$$J_{[]}\left(\delta_n, \mathcal{F}_n, d\right) \ \lesssim \ \delta_n^2 \sqrt{n}.$$

Cost of the union? To be defined.

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Proof of Proposition 1

Recall that the oracle is given by $D_\theta = \dfrac{p_0}{p_0 + p_\theta}$, hence $\dfrac{p_0}{p_\theta} = \dfrac{D_\theta}{1 - D_\theta}$.

- u_θ vanishes at the oracle.
- Under the overlap assumption (p_0 and p_θ put mass on the same region of X), the orthogonalization correction vanishes:

$$P_\theta\Big[\frac{D_\theta}{1 - D_\theta}\Big] = \int_{p_\theta > 0} \frac{p_0}{p_\theta}\, p_\theta d\mu = \int_{p_\theta > 0} p_0 d\mu = 1 \qquad \Longrightarrow \qquad \widetilde{L} = \widetilde{L} + (1 - 1) = \widetilde{L}.$$

- Therefore

$$\widetilde{L}(\theta) = L(\theta) = -D_{\mathrm{KL}}\big(p_0 \,\|\, p_\theta\big).$$

□

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Proof of Lemma 1

- Let $C_0 = \sum_i \log(p_0(X_i))$. The classification-based likelihood can be writen as

$$\begin{aligned} n\widetilde{L}_n(\theta) &= nL_n(\theta) + u_\theta(X^{(n)}) \\ &= C_0 + \sum_{i=1}^{n} \log \frac{1 - D_\theta(X_i)}{D_\theta(X_i)} + \sum_{i=1}^{n} \left[\log \frac{1 - \widehat{D}^\theta_{n,m}(X_i)}{1 - D_\theta(X_i)} - \log \frac{\widehat{D}^\theta_{n,m}(X_i)}{D_\theta(X_i)} \right] \\ &= C_0 + \sum_{i=1}^{n} \left[\log \left(1 - \widehat{D}^\theta_{n,m}(X_i)\right) - \log \widehat{D}^\theta_{n,m}(X_i) \right], \end{aligned}$$

  No intractable term survives in the difference.
- The orthogonalization term depends only on $\widehat{D}^\theta_{n,m}$.

□

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Proof of Lemma 2

- We know that $p_\theta = p_0 \frac{1-D_\theta}{D_\theta} =: p^{D_\theta}$. So $p^{D_\theta} = p_\theta$ and $p^{1/2} = p_0$.
- We also know that $d(D, D') := h(p^D, p^{D'})$, therfore

$$d\big(D_\theta, \tfrac{1}{2}\big) = h(p_\theta, p_0) \qquad \text{for every } \theta \in \Theta.$$

- It follows that
  (i) $\theta \in \Theta_I \Rightarrow p_\theta = p_0 \Rightarrow h = 0 \Rightarrow D_\theta = \frac{1}{2}$, by the first part of Lemma 2.
  (ii) $\theta \in \Theta_{osn} \Rightarrow h(p_\theta, p_0) < \varepsilon_n$, by Definition 1.

□

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Algorithm

Input: N particles drawn from the prior ($\gamma = 0$) and the criterion $\widetilde{L}_n$ evaluated on it.

1. **Raise the temperature slowly** from γ to γ'.
2. **Reweight** each particle by $\exp\left((\gamma' - \gamma)\, n\widetilde{L}_n(\theta^{(b)})\right)$, then normalise.
3. **Move** each particle: propose a nearby θ', evaluate $\widetilde{L}_n$ at both, and accept with the **tempered** ratio

$$\min\left\{ \exp\left(\gamma'\, n\left[\widetilde{L}_n(\theta') - \widetilde{L}_n(\theta)\right]\right) \frac{\pi(\theta')}{\pi(\theta)} \frac{q(\theta \mid \theta')}{q(\theta' \mid \theta)},\ 1 \right\}$$

4. Repeat until $\gamma = 1$: the particles are then draws from the **quasi-posterior**.

Output: $\{\theta^{(b)}\}_{b=1}^{N} \sim \widehat{\Pi}_n$

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